Northstar AI Platform

Enterprise Operations, Security, Reliability, and Data Governance Manual

Purpose. This synthetic manual was created as a production-style benchmark for retrieval, compression, provenance, and evidence-preservation systems. It deliberately mixes critical requirements with plausible but irrelevant operational details.

Document Control

Owner	Platform Engineering
Classification	Internal - Benchmark
Revision	3.4
Effective date	September 1, 2026
Review cycle	Quarterly

The platform supports internal analysts, customer-facing applications, and automated batch workloads. The manual is intentionally verbose: some paragraphs describe office routines, documentation conventions, or historical choices that are not relevant to technical extraction tasks. Other paragraphs contain exact thresholds, deadlines, limits, and mandatory controls that should be preserved by a reliable context compiler.

A successful retrieval system should preserve task-relevant facts even when they are separated by unrelated prose, expressed with numbers, or embedded in sections whose headings carry important context. It should also avoid selecting superficially similar distractors such as meeting times, cafeteria schedules, design preferences, or unrelated dashboard refresh intervals.

1. Identity and Authentication

Authentication Architecture

Northstar uses centrally issued JSON Web Tokens for authenticated API access. **Authentication requires JWT access tokens.** The tokens are signed by the identity service and validated by gateway middleware before a request is forwarded to application services.

Token Lifetimes

Access tokens expire after 15 minutes. Refresh tokens expire after **7 days** and must be rotated after use. Token validation must reject expired signatures even when the user still has an active browser session. Administrators should not extend token lifetimes to work around client bugs.

Administrative Access

MFA is required for administrator accounts. Password reset requires identity verification through the approved service desk workflow. Emergency break-glass accounts are reviewed after every use and may not be used for routine administration.

Login Protection

After five failed login attempts, the account is locked for 20 minutes. Repeated authentication failures are sent to the security event stream. A lockout may be manually cleared only after the user identity is confirmed.

Distractor Notes

The office cafeteria opens at 7:00 AM on weekdays. Visitor parking permits are issued at the front desk, and conference rooms may be reserved up to 30 days in advance. These facilities rules are unrelated to authentication enforcement.

2. API Deployment and Request Handling

Production Endpoints

The production API listens on port 8001. Health probes use the **/api/health** endpoint. The development web interface normally runs on port 3000, but that frontend port is not part of the production API contract.

Request Limits

Requests larger than 25 MB are rejected. Payload limits apply before parsing so oversized uploads do not consume excessive memory. File type checks occur after the request passes the size gate.

Transient Failures

For retryable upstream failures, **the service allows three retry attempts** with bounded exponential backoff. Authentication failures, schema validation errors, and permission denials are not retried automatically.

Operational Detail

The platform team holds a 9:30 AM stand-up three days per week. Engineers may use dark or light editor themes. These preferences have no effect on API availability or retry policy.

Deployment Safety

Production releases require a successful health check, migration verification, and rollback readiness. A release is considered healthy only after error rate and latency remain within their alerting bands for the observation window.

3. Database Runtime and Backup Policy

Connection Pool

The database pool is limited to 30 connections. Pool exhaustion should return a controlled service error instead of spawning unbounded connections. Long-lived administrative sessions count toward the same limit.

Slow Query Handling

Queries exceeding 5 seconds are logged as slow queries. Slow-query events include the normalized statement identifier, duration, database role, and request correlation ID. Parameter values containing secrets must not be logged.

Connection Retry

Failed database connections retry three times before raising an error. Retry logic applies only to transient connectivity failures. Authentication failures and malformed connection strings fail immediately.

Backup Schedule

Backups run every night at 2 AM. Backup retention is **30 days**. A restore test is performed on the first Monday of each month, and failures are tracked as reliability incidents.

Distractor Operations

The marketing dashboard refreshes every 45 minutes. The team sometimes orders pizza on Fridays, and the office coffee machine was replaced last quarter. Design assets are archived yearly. None of these activities alter database backup or connection policy.

4. Data Protection and Retention

Encryption

Customer records must be encrypted at rest using AES-256. Encryption keys are stored in the managed key service and rotated according to the key-management schedule. Application logs must not contain raw encryption keys or decrypted secrets.

Export Restrictions

Production database exports cannot be stored on personal devices. Approved exports must use managed storage with access logging and automatic expiration. Temporary local copies created by administrative tooling must be deleted immediately after the approved transfer completes.

Access Approval

Access to customer records requires manager approval and MFA. Approval records must identify the requestor, business purpose, approving manager, and expiration date of the access grant.

Retention

Audit logs must be retained for a minimum of 365 days. Deleted customer records remain in encrypted backups for **no more than 30 days**. Legal holds may extend retention only when formally authorized.

Distractor Benefits

Employees receive a monthly wellness newsletter. Parking permits are renewed each January, and furniture requests are handled through the facilities portal. These administrative processes are intentionally unrelated to customer-data controls.

5. Incident Response and Severity Management

Severity Classification

A production outage affecting authentication is classified as Severity 1. Severity is determined by customer impact, security exposure, and breadth of service degradation, not by the number of engineers currently available.

Paging

Severity 1 incidents require paging the on-call engineer immediately. The page must include the affected service, initial symptoms, and any known customer impact.

Command Structure

The incident commander must be assigned within 10 minutes. The commander coordinates technical response, communications, and escalation. A separate scribe maintains the event timeline.

Customer Communication

Customer-facing status updates must be published every 30 minutes until mitigation. Updates should state what is known, what remains uncertain, and the current mitigation status without speculating about unverified root causes.

Post-Incident

A written postmortem is required within five business days for every Severity 1 incident. Action items must have an owner and due date. The support team rotates snack purchases weekly, which is unrelated to incident response.

6. Observability, Logging, and Alerting

Service-Level Signals

Each production service emits latency, error-rate, throughput, and saturation metrics. The primary API latency objective is measured at the 95th percentile so isolated outliers do not dominate the service-level dashboard.

Alert Thresholds

An alert is triggered when the five-minute error rate exceeds 2.0% for two consecutive evaluation windows. A latency alert is triggered when p95 response time remains above **850 ms for 10 minutes**. Warning-only events do not page the on-call engineer unless they coincide with customer-visible degradation.

Log Hygiene

Application logs must include a correlation ID, service name, environment, and severity. Secrets, raw access tokens, passwords, and full payment-card values must never be written to logs.

Metrics Retention

High-resolution metrics are retained for 14 days before downsampling. Aggregated metrics are retained for 13 months to support seasonal capacity analysis.

Distractor Reporting

The executive dashboard refreshes every 60 minutes, and the weekly operations review begins at 3 PM on Thursdays. These reporting schedules are not alert thresholds.

7. Privacy and Sensitive Data Handling

Sensitive Fields

Personally identifiable information includes full names paired with account identifiers, government-issued identifiers, precise home addresses, and unmasked financial account numbers. Service owners must document which fields their systems collect.

Production Access

Direct production access to unmasked PII is limited to the Privacy Operations and Security Response roles. Engineers investigating routine defects should use masked or synthetic data whenever possible.

Deletion Requests

Verified deletion requests must be completed within 30 calendar days. The completion record must include the request identifier, verification status, systems affected, and completion timestamp.

Analytics

Analytics datasets must remove direct identifiers before analyst access is granted. Re-identification attempts are prohibited and treated as a policy violation.

Distractor Culture

The privacy team hosts an optional lunch-and-learn once per quarter. Attendance is encouraged but is not a substitute for mandatory access-control or deletion requirements.

8. Embedding, Cache, and Model Runtime

Embedding Cache

Embedding results are stored in a persistent SQLite cache. The current production schema identifier is **v3**. Cache entries are keyed by the normalized input hash, model identifier, and relevant preprocessing version.

Model Runtime

The default local embedding model is all-MiniLM-L6-v2. Model downloads may be served from a shared artifact cache so repeated deployments do not fetch identical weights from the network.

Cache Integrity

Corrupted cache entries must be discarded and recomputed rather than returned to callers. Cache health metrics include hit rate, write failures, schema mismatches, and database size.

Capacity

The cache is compacted when the SQLite file exceeds 2 GB. Compaction may be deferred during peak request periods to avoid unnecessary I/O contention.

Distractor Development Notes

The frontend logo was updated during a design review, and developers may choose their preferred editor extensions. These facts do not affect cache behavior or embedding correctness.

9. Change Management and Release Controls

Approval

Production changes require at least one approving reviewer who did not author the change. Emergency changes may use the expedited process but still require retrospective review within one business day.

Rollback

Every production deployment must have a documented rollback procedure before release begins. A rollback is preferred over live debugging when a release causes sustained customer impact and the rollback path is known to be safe.

Database Migrations

Destructive schema migrations require a verified backup less than 24 hours old. Migration scripts must be idempotent where practical and tested against a production-like dataset before approval.

Deployment Windows

Routine releases are normally scheduled during lower-traffic periods, but the schedule itself does not waive approval, backup, or rollback requirements. The team calendar also includes social events that are unrelated to release governance.

10. Third-Party Services and Vendor Risk

Risk Review

Vendors that process customer data must complete a security review before production use. The review evaluates encryption, access control, incident notification, data location, subprocessor use, and deletion capability.

Incident Notification

Critical vendors must notify Northstar of a confirmed security incident within 24 hours. Contracts should preserve the right to request evidence of remediation and relevant forensic findings.

Data Location

Customer production data may be stored only in approved U.S. regions unless Legal and Privacy jointly approve an exception. Sandbox datasets may use broader regions only when they contain no customer production data.

Annual Review

Critical vendors are reassessed at least once every 12 months. The procurement system also tracks billing contacts and renewal dates, but those commercial fields are not part of the technical security controls.

11. Disaster Recovery and Business Continuity

Recovery Objectives

The production platform recovery time objective (RTO) is 4 hours. The recovery point objective (RPO) for transactional customer data is **15 minutes**. Services that cannot meet these objectives must document an approved exception.

Regional Failure

If the primary region is unavailable for more than 20 minutes, the incident commander may authorize regional failover. Failover requires confirmation that replication lag is within the RPO target and that dependent services are reachable.

Exercises

A full disaster-recovery exercise is conducted twice per year. Exercise reports must document recovery duration, data-loss observations, communication gaps, and corrective actions.

Continuity Distractors

The company picnic is usually held in June, and office inventory is reconciled each December. These events are unrelated to disaster-recovery objectives and failover policy.

> **Benchmark Note**
>
> This document intentionally distributes critical evidence across many pages while including realistic distractors, repeated concepts, and unrelated numeric facts. It is suitable for testing task-aware retrieval, provenance, compression, and evidence preservation.